A MONOLITHIC SOLVER FOR LINEARISED NAVIER–STOKES FREE-SURFACE DYNAMICS VIA HELMHOLTZ DECOMPOSITION

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[Author 1]*Department>Department
University>University

[email]

[Author 2] Department>Department
University>University

[email]

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ABSTRACT

We present a numerical implementation of the linearised Navier–Stokes (LNS) free-surface model of **?**, which employs a Helmholtz decomposition of the velocity field into irrotational and vortical components. The model solves the coupled system of Laplace's equation for the velocity potential, heat equations for the vortical fields, and surface evolution equations that incorporate gravity, surface tension, and viscous effects. We implement both a sequential (operator-splitting) solver and a monolithic formulation in which all unknowns—the vortical components w^3 and w^1, the surface potential ϕ_s, the bulk potential ϕ, and the surface elevation η—are assembled into a single linear system $A\mathbf{x} = \mathbf{b}$. Because the coefficient matrix A is independent of the solution, it is factored once via sparse LU decomposition; each subsequent time step requires only $\mathcal{O}(N)$ back-substitution. We verify that the two formulations produce identical results to machine precision and demonstrate the computational advantage of the monolithic approach. We further extend the same monolithic framework to fluid–body coupling by augmenting the unknown vector with the surface pressure, vertical position, and vertical velocity of a floating rigid body, yielding a fully implicit solver in which the body motion is determined self-consistently by Newton's second law driven by gravity and the integrated fluid pressure.

Keywords Free surface dynamics · linearised Navier–Stokes · Helmholtz decomposition · monolithic solver · LU prefactorisation · fluid–body coupling · floating rigid body · finite differences

1 Introduction

The dynamics of viscous free-surface flows arise in numerous physical settings, from ocean wave propagation and droplet impact to the interaction of floating bodies with fluid baths. While the full Navier–Stokes equations govern such flows, their nonlinearity and the need to track a moving free surface make direct numerical solution computationally expensive. For problems where surface deflections are small relative to the fluid depth, a linearised approach can retain the essential physics—gravity, surface tension, and viscous dissipation—at greatly reduced cost.

? developed such a linearised Navier–Stokes (LNS) framework in the context of non-wetting sphere impact on a fluid bath and bouncing droplet dynamics. The central idea is a Helmholtz decomposition of the velocity field $\mathbf{u} = \nabla\phi + \mathbf{w}$, which separates the flow into an irrotational component $\nabla\phi$ satisfying Laplace's equation in the bulk and a vortical component $\mathbf{w}$ governed by the heat equation. The two components are coupled through linearised boundary conditions at the free surface, which encode the kinematic condition (the surface moves with the fluid), the dynamic condition (normal stress balance including gravity, surface tension, and viscous effects), and a vorticity evolution equation. At the bottom boundary, the non-slip condition $\mathbf{u} = 0$ couples the vortical field to the potential field.

*Corresponding author.

This framework was subsequently used by **?** to study Faraday pilot-wave dynamics, by **?** to investigate hydrodynamic pilot-wave resonance, and more recently by **?** in the context of macroscopic Brownian motion on chaotic fluid interfaces. **?** applied the framework to the impact of a rigid sphere onto an elastic membrane, demonstrating its applicability to fluid–structure interaction problems. The classical potential-flow foundations trace back to **?** and **?**.

A common simplification employed in some implementations is the diagnostic relation $w^3 = (2/\mathrm{Re})\,\Delta_H \eta$ at the surface, derived by combining the kinematic and vorticity surface equations under the assumption that initial vorticity transients have decayed. While this eliminates one unknown and is valid at high Reynolds numbers, it does not capture the full transient behaviour of the surface vorticity, and it precludes correct treatment of the non-slip bottom boundary condition for the vortical field. In the present work, we solve the *full* set of equations (2.15a–e) from **?**, evolving w^3 and w^1 as independent 2D fields via their respective heat equations, with proper boundary conditions at both the free surface and the solid bottom.

Our main contribution is a monolithic formulation of the full system. In the standard sequential (operator-splitting) approach, each time step involves multiple sparse solves: the heat equations for w^3 and w^1, the implicit surface update for ϕ_s, the Laplace equation for the bulk potential, and the kinematic update for η. We show that the entire system can be assembled into a single block-structured linear system

$$A\,\mathbf{x}^{n+1} = \mathbf{b}(\mathbf{x}^n),$$

where $\mathbf{x} = [w^3,\, w^1,\, \phi_s,\, \phi_{\mathrm{bulk}},\, \eta]^\top$. Because all equations are linear with constant coefficients, the matrix A does not depend on the solution. It is factored once via sparse LU decomposition at setup time, and each time step reduces to a single back-substitution at $\mathcal{O}(N)$ cost. We verify that the monolithic and sequential solvers produce identical results to machine precision, confirming the equivalence of the two formulations.

We then extend this framework to fluid–body coupling. A flat-bottomed rigid body of finite width is placed on the free surface, and its motion is determined self-consistently by Newton's second law driven by gravity and the integrated fluid pressure. Under the wetted region of the body, the standard free-surface boundary conditions are replaced by three constraints: a geometric constraint (η matches the body position), a kinematic match (the vertical fluid velocity matches the body velocity), and horizontal no-slip at the body bottom. The surface pressure p_s acts as a Lagrange multiplier enforcing the kinematic match. Augmenting the unknown vector by p_s and the two body scalars z_b, v_b, the entire coupled system remains linear with constant coefficients, so it too is amenable to a one-time LU prefactorisation. This fully implicit coupling avoids the added-mass instability that is known to afflict staggered fluid–body coupling schemes, particularly for light bodies.

The remainder of this paper is organised as follows. Section **??** presents the mathematical model: the Helmholtz decomposition, the governing equations in the bulk and at the boundaries, the non-dimensionalisation, and the extension to fluid–body coupling for a floating rigid body. Section **??** describes the numerical methods: finite difference discretisation, the sequential time-stepping algorithm, the monolithic system assembly for the pure fluid problem, and the augmented monolithic system for the coupled fluid–body problem. Section **??** presents numerical results.

2 Mathematical model

2.1 Physical setup

We consider a two-dimensional domain with horizontal coordinate $x' \in [0, L']$ and vertical coordinate z', where $z' = 0$ is at the mean free surface and $z' = -D'$ is at the solid bottom. The fluid has density ρ, kinematic viscosity ν, and surface tension coefficient σ. The free surface is located at $z' = \eta'(x', t')$, where η' is the surface elevation. Primes denote dimensional variables.

We assume that surface deflections are small relative to the depth ($|\eta'| \ll D'$), so that the free-surface boundary conditions can be linearised about $z' = 0$. Horizontal boundary conditions are periodic: $\eta'(0, t') = \eta'(L', t')$, and similarly for all other fields.

2.2 Helmholtz decomposition

Following **?**, the velocity field is decomposed as

$$\mathbf{u} = \nabla\phi + \mathbf{w}, \tag{1}$$

where $\phi(x', z', t')$ is the velocity potential and $\mathbf{w}(x', z', t')$ is the vortical component. Both fields are solenoidal: $\Delta\phi = 0$ (by construction) and $\nabla \cdot \mathbf{w} = 0$ (from incompressibility). In two dimensions, $\mathbf{w} = (w^1, w^3)$ has a horizontal component w^1 and a vertical component w^3; the out-of-plane component $w^2 = 0$ by symmetry.

2.3 Governing equations

2.3.1 Bulk equations

In the interior of the fluid domain ($z' < 0$), the potential satisfies Laplace's equation and each vortical component satisfies the heat equation:

$$\Delta\phi = 0, \tag{2}$$

$$\partial_{t'} w'^i = \nu\, \Delta w'^i, \quad i = 1, 3. \tag{3}$$

2.3.2 Free-surface boundary conditions ($z' = 0$)

The linearised boundary conditions at the free surface are (cf. **?**, equations 2.15c–e):

Kinematic condition (2.15c): The surface moves with the fluid:

$$\partial_{t'} \eta' = \partial_{z'} \phi' + w'^3. \tag{4}$$

Dynamic condition (2.15d): The normal stress balance at the free surface:

$$\partial_{t'} \phi'_s = -g\eta' + \frac{\sigma}{\rho}\, \partial_{x'x'} \eta' + 2\nu\, \partial_{x'x'} \phi'_s - \frac{2\nu\, \partial_{z'} w'^3}{} - \frac{p'_s}{\rho}, \tag{5}$$

where $\phi'_s(x', t') = \phi'(x', 0, t')$ is the surface velocity potential, $\partial_{x'x'}$ is the horizontal Laplacian (which equals Δ_H in 2D), p'_s is an externally applied surface pressure (zero in our case), and the curvature $\kappa[\eta'] \approx \partial_{x'x'} \eta'$ under the linearisation assumption. Note the term $-2\nu\, \partial_{z'} w'^3$, which represents the viscous contribution from the vertical gradient of the vortical field.

Vorticity evolution (2.15e): The surface vorticity evolves according to:

$$\partial_{t'} w'^3 = 2\nu\, \partial_{x'x'} \left(\partial_{z'} \phi' + w'^3 \right). \tag{6}$$

2.3.3 Bottom boundary condition ($z' = -D'$)

The non-slip condition $\mathbf{u} = 0$ at the solid bottom requires, via the decomposition (**??**):

$$w'^3 = -\partial_{z'} \phi' \quad \text{at } z' = -D', \tag{7}$$

$$w'^1 = -\partial_{x'} \phi' \quad \text{at } z' = -D'. \tag{8}$$

2.3.4 Stress-free condition for w^1 at the surface

The tangential stress-free condition at the free surface (cf. **?**, equation 2.11a) gives:

$$\partial_{z'} w'^1 = -\left(\partial_{x'z'} \phi' + \partial_{x'} w'^3 \right) \quad \text{at } z' = 0. \tag{9}$$

Remark on equation (2.17). Combining (**??**) and (**??**) and integrating in time, one can derive the diagnostic relation $w^3 = (2\nu/\text{etc.})\, \Delta_H \eta$ at the surface (**?**). This is valid after initial transients have decayed and at high Reynolds numbers. In the present implementation, we do *not* use this simplification; instead, we solve equation (**??**) directly, which correctly captures transient vorticity dynamics and is consistent with the non-slip bottom boundary condition.

2.4 Non-dimensionalisation

We non-dimensionalise using a characteristic length L_c and velocity $V_c = L_c/T_c$:

$$x = \frac{x'}{L_c}, \quad z = \frac{z'}{L_c}, \quad t = \frac{t'}{T_c}, \quad \eta = \frac{\eta'}{L_c}, \quad \phi = \frac{\phi'}{V_c L_c}. \tag{10}$$

This introduces the dimensionless numbers

$$\mathrm{Fr} = \frac{V_c^2}{g L_c}, \qquad \mathrm{We} = \frac{\rho\, V_c^2 L_c}{\sigma}, \qquad \mathrm{Re} = \frac{V_c L_c}{\nu}. \tag{11}$$

The dimensionless governing equations become:

Bulk:

$$\Delta\phi = 0, \quad z \le 0, \tag{12}$$

$$\partial_t w^i = \frac{1}{\mathrm{Re}}\,\Delta w^i, \quad z \le 0, \quad i = 1, 3. \tag{13}$$

Free surface ($z = 0$):

$$\partial_t \eta = \partial_z \phi + w^3, \tag{14}$$

$$\partial_t \phi_s = -\frac{1}{\mathrm{Fr}}\,\eta + \frac{1}{\mathrm{We}}\,\partial_{xx}\eta + \frac{2}{\mathrm{Re}}\,\partial_{xx}\phi_s - \frac{2}{\mathrm{Re}}\,\partial_z w^3, \tag{15}$$

$$\partial_t w^3 = \frac{2}{\mathrm{Re}}\,\partial_{xx}\big(\partial_z\phi + w^3\big)\,. \tag{16}$$

Bottom ($z = -D$):

$$w^3 = -\partial_z \phi, \tag{17}$$

$$w^1 = -\partial_x \phi. \tag{18}$$

Surface stress-free for w^1:

$$\partial_z w^1 = -(\partial_{xz}\phi + \partial_x w^3) \quad \text{at } z = 0. \tag{19}$$

2.5 Extension to a floating rigid body

The LNS framework extends naturally to the case of a rigid body floating on the fluid surface. We consider a flat-bottomed rectangular body of half-width a and mass per unit length m, centred at $x = x_c$, whose bottom lies at dimensional vertical position $z_b'(t')$ with vertical velocity $v_b'(t') = \mathrm{d}z_b'/\mathrm{d}t'$. At $t' = 0$ the body is in contact with the fluid surface. Unlike prescribed-motion formulations, the body motion here is determined self-consistently by Newton's second law driven by gravity and the integrated fluid pressure.

Body equations of motion. In dimensionless form, with $M = m/(\rho L_c^2)$ denoting the dimensionless mass per unit length,

$$\frac{\mathrm{d}z_b}{\mathrm{d}t} = v_b, \tag{20}$$

$$M\,\frac{\mathrm{d}v_b}{\mathrm{d}t} = -\frac{M}{\mathrm{Fr}} + \int_{|x-x_c|<a} p_s(x,t)\,\mathrm{d}x, \tag{21}$$

where p_s is the surface pressure. The integral in (**??**) represents the net upward force exerted by the fluid on the wetted region of the body.

Boundary conditions under the wetted region. Let $\mathcal{B}(t) = \{x : |x - x_c| < a\}$ denote the wetted region. Under $\mathcal{B}$, the standard free-surface boundary conditions are replaced by three conditions imposed by the presence of the solid body:

1. *Surface follows the body* (geometric constraint):
$$\eta(x,t) = z_b(t), \qquad x \in \mathcal{B}(t). \tag{22}$$
2. *Kinematic match* (no-penetration at the body bottom):
$$\partial_z\phi + w^3 = v_b(t), \qquad x \in \mathcal{B}(t),\ z = 0. \tag{23}$$
This replaces the free-surface kinematic condition (**??**) under the body; its compatibility with (**??**) follows from $\mathrm{d}z_b/\mathrm{d}t = v_b$.
3. *Horizontal no-slip* at the body bottom:
$$\partial_x\phi + w^1 = 0, \qquad x \in \mathcal{B}(t),\ z = 0. \tag{24}$$
This replaces the surface stress-free condition (**??**) under the body.

The dynamic boundary condition (**??**) continues to hold at $z = 0$ everywhere, but under $\mathcal{B}$ the surface pressure p_s is no longer prescribed to zero—it becomes an unknown whose role is that of a Lagrange multiplier enforcing the kinematic match (**??**). Outside the wetted region, $p_s = 0$ and the standard free-surface boundary conditions (**??**)–(**??**) and (**??**) apply.

Summary of the coupled problem. The full coupled fluid–body problem thus consists of the bulk fluid equations (**??**)–(**??**), the piecewise-defined surface boundary conditions summarised in Table **??**, the body equations of motion (**??**)–(**??**), and the bottom non-slip conditions (**??**)–(**??**). The additional unknowns introduced by the body are the surface pressure field $p_s(x,t)$ (supported only on $\mathcal{B}$), together with the two scalar body states $z_b(t)$ and $v_b(t)$.

Table 1: Surface boundary conditions as a function of region. The dynamic condition (**??**) applies uniformly, but with p_s prescribed to zero outside the body and left as an unknown inside.

Region	Free surface ($x \notin \mathcal{B}$)	Wetted body ($x \in \mathcal{B}$)
Surface elevation	$\partial_t \eta = \partial_z \phi + w^3$	$\eta = z_b$
Kinematic	(evolves η)	$\partial_z \phi + w^3 = v_b$
Horizontal vortical w^1	$\partial_z w^1 = -(\partial_{xz}\phi + \partial_x w^3)$	$\partial_x \phi + w^1 = 0$
Surface pressure	$p_s = 0$	p_s unknown

3 Numerical methods

3.1 Spatial discretisation

We discretise the domain on a uniform rectangular grid with n_x points in x and n_z points in z:

$$x_j = j\,\Delta x, \quad j = 0, \ldots, n_x - 1; \qquad z_i = -D + i\,\Delta z, \quad i = 0, \ldots, n_z - 1, \tag{25}$$

with $\Delta x = L/n_x$ and $\Delta z = D/(n_z - 1)$. The index $i = 0$ corresponds to the bottom ($z = -D$) and $i = n_z - 1$ to the free surface ($z = 0$).

Standard second-order finite differences are used throughout:

- **Horizontal Laplacian** $\Delta_H = \partial_{xx}$: three-point stencil with periodic wrapping,
$$(\Delta_H u)_j = \frac{u_{j+1} - 2u_j + u_{j-1}}{(\Delta x)^2}, \quad j \pm 1 \text{ taken modulo } n_x. \tag{26}$$
- **Vertical derivative**: centred differences in the interior, one-sided at boundaries.
- **2D Laplacian**: standard five-point stencil $\Delta = \Delta_H + \partial_{zz}$.

3.2 Sequential solver

The sequential solver advances the solution from time level n to $n+1$ in four sub-steps, each involving one or more sparse linear solves.

Step 1: Evolve w^3 and w^1 (heat equations). The bulk heat equation (**??**) is discretised implicitly:

$$\left(I - \frac{\Delta t}{\mathrm{Re}}\,\Delta_{2\mathrm{D}}\right) w^{i,n+1} = w^{i,n}, \quad i = 1, 3, \tag{27}$$

where $\Delta_{2\mathrm{D}}$ is the discrete 2D Laplacian. After solving the bulk, the boundary conditions are overwritten:

- *Bottom* ($i = 0$): $w_j^{3,n+1} = -(\phi_{1,j}^n - \phi_{0,j}^n)/\Delta z$ and $w_j^{1,n+1} = -(D_x\,\phi_0^n)_j$, where D_x is the centred difference operator in x.
- *Surface* ($i = n_z - 1$) *for* w^3: equation (**??**) discretised implicitly,
$$\left(I - \frac{2\Delta t}{\mathrm{Re}}\,\Delta_H\right) w_s^{3,n+1} = w_s^{3,n} + \frac{2\Delta t}{\mathrm{Re}}\,\Delta_H\,\phi_z^n, \tag{28}$$
where $\phi_z^n = (\phi_{n_z-1}^n - \phi_{n_z-2}^n)/\Delta z$.
- *Surface for* w^1: stress-free condition (**??**) discretised as $w_{n_z-1}^1 = w_{n_z-2}^1 - \Delta z\,(\phi_{xz}^n + D_x\,w_s^{3,n+1})$.

Step 2: Update surface potential ϕ_s (dynamic condition). Equation (**??**) is discretised implicitly in the diffusion term:

$$\left(I - \frac{2\Delta t}{\mathrm{Re}}\,\Delta_H\right) \phi_s^{n+1} = \phi_s^n - \frac{\Delta t}{\mathrm{Fr}}\,\eta^n + \frac{\Delta t}{\mathrm{We}}\,\kappa[\eta^n] - \frac{2\Delta t}{\mathrm{Re}}\,(D_z\,w^{3,n+1})_s, \tag{29}$$

where $\kappa[\eta] = \Delta_H \eta$ (linearised curvature) and $(D_z\,w^3)_s = (w_{n_z-1}^3 - w_{n_z-2}^3)/\Delta z$.

Step 3: Solve Laplace equation for bulk ϕ. With the updated surface Dirichlet condition $\phi_{n_z-1,j} = (\phi_s)_j^{n+1}$ and the bottom ghost-point condition enforcing $\partial_z \phi = -w_0^{3,n+1}$ (from (**??**)), we solve:

$$A_{\text{Lap}}\, \vec{\phi}^{n+1} = \vec{b}_{\text{Lap}}, \tag{30}$$

where A_{Lap} encodes the five-point Laplacian with periodic horizontal and appropriate vertical boundary conditions.

Step 4: Update surface elevation η (kinematic condition). Equation (**??**) is updated explicitly:

$$\eta^{n+1} = \eta^n + \Delta t \, \left(\phi_z^{n+1} + w_s^{3,n+1}\right), \tag{31}$$

where $\phi_z^{n+1} = (\phi_{n_z-1}^{n+1} - \phi_{n_z-2}^{n+1})/\Delta z$.

Each time step thus requires solving: two $n_z n_x$-sized heat systems (for w^3 and w^1), two n_x-sized tridiagonal-like surface systems (for w_s^3 and ϕ_s), and one $n_z n_x$-sized Laplace system.

3.3 Monolithic formulation

The sequential solver calls `spsolve` multiple times per step. We now show that the entire system can be assembled into a single matrix equation.

3.3.1 Unknown vector

The full state at time level $n+1$ is collected into a single vector:

$$\mathbf{x} = \begin{pmatrix} \vec{w}^3 \\ \vec{w}^1 \\ \vec{\phi}_s \\ \vec{\phi}_{\text{bulk}} \\ \vec{\eta} \end{pmatrix}, \qquad N_{\text{total}} = 3\, n_z n_x + 2\, n_x. \tag{32}$$

3.3.2 Block structure

The monolithic system $A\,\mathbf{x}^{n+1} = \mathbf{b}(\mathbf{x}^n)$ has a block lower-triangular structure:

Row block	w^3	w^1	ϕ_s	ϕ_{bulk}	η	Equation
w^3	•					Heat eq. + BCs
w^1	•	•				Heat eq. + BCs
ϕ_s	•		•			Dynamic BC (**??**)
ϕ_{bulk}	•		•	•		Laplace eq.
η	•			•	•	Kinematic BC (**??**)

The block lower-triangular structure arises because:

- The w^3 block depends only on $w^{3,n+1}$ (implicitly) and ϕ^n (explicitly, in the RHS).
- The w^1 block depends on $w^{1,n+1}$ and $w^{3,n+1}$ (through the stress-free surface BC).
- The ϕ_s block depends on ϕ_s^{n+1} and $w^{3,n+1}$ (through $\partial_z w^3$).
- The Laplace block depends on ϕ_{bulk}^{n+1}, ϕ_s^{n+1} (Dirichlet top), and $w^{3,n+1}$ (ghost point at bottom).
- The η block depends on η^{n+1}, ϕ_{bulk}^{n+1} (through ϕ_z), and $w_s^{3,n+1}$.

3.3.3 Key property: constant coefficient matrix

Since all governing equations are linear with constant coefficients, the matrix A depends only on grid parameters (n_x, n_z, Δx, Δz), the time step Δt, and the dimensionless numbers (Fr, We, Re)—not on the solution. Therefore, A is factored once at setup time via sparse LU decomposition:

$$A = LU \quad \text{(computed once)}. \tag{33}$$

At each time step, the right-hand side $\mathbf{b}(\mathbf{x}^n)$ is assembled from the current state, and the new state is obtained by back-substitution:

$$\mathbf{x}^{n+1} = U^{-1}(L^{-1}\mathbf{b}(\mathbf{x}^n)) \quad (\mathcal{O}(N) \text{ per step}). \tag{34}$$

For our grid ($n_x = 300$, $n_z = 30$), the monolithic system has $N_{\text{total}} = 27{,}600$ unknowns with approximately $134{,}400$ nonzero entries. The LU factorisation takes approximately 0.06 s and each back-substitution approximately 1.9 ms, compared to the sequential solver which requires approximately 34 ms per step.

3.4 Augmented monolithic system with body coupling

To solve the coupled fluid–body problem described in Section **??**, we extend the monolithic approach of Section **??** by augmenting the unknown vector with the surface pressure field and the two scalar body states. Remarkably, the resulting system retains the same property as the pure fluid system: the coefficient matrix is constant in time and can be factored once via LU decomposition, after which each time step costs only a single back-substitution.

3.4.1 Augmented unknown vector

At time level $n+1$, the augmented state vector is

$$\mathbf{x} = \begin{pmatrix} \vec{w}^3 \\ \vec{w}^1 \\ \vec{\phi} \\ \vec{\eta} \\ \vec{p}_s \\ z_b \\ v_b \end{pmatrix}, \qquad N_{\text{total}} = 3\,n_z n_x + 2\,n_x + 2, \tag{35}$$

where we have replaced the separate surface and bulk potential vectors of (**??**) with a single ϕ vector covering all grid points (the surface row at $i = n_z - 1$ carries the values formerly denoted ϕ_s), and introduced three new components: the surface pressure $\vec{p}_s \in \mathbb{R}^{n_x}$ (with $p_{s,j} = 0$ enforced for $j \notin \mathcal{B}$), and the scalar body states $z_b, v_b \in \mathbb{R}$.

3.4.2 Piecewise surface boundary conditions

Let $\chi_j = 1$ if node $j \in \mathcal{B}$ and zero otherwise. The surface rows of the monolithic matrix differ between free and wetted nodes according to Table **??**:

η rows. For free j: the discretised kinematic condition (**??**),

$$\eta_j^{n+1} - \frac{\Delta t}{\Delta z}\left(\phi_{n_z-1,j}^{n+1} - \phi_{n_z-2,j}^{n+1}\right) - \Delta t\, w_{n_z-1,j}^{3,n+1} = \eta_j^n. \tag{36}$$

For body j: the geometric constraint (**??**),

$$\eta_j^{n+1} - z_b^{n+1} = 0. \tag{37}$$

w^1 surface rows. For free j: the stress-free condition (**??**). For body j: the horizontal no-slip condition (**??**),

$$w_{n_z-1,j}^{1,n+1} + \left(D_x\,\vec{\phi}_{n_z-1}^{n+1}\right)_j = 0, \tag{38}$$

which couples the surface vortical component to the surface potential via the discrete first-derivative operator D_x.

p_s rows. For free j: the prescribed value

$$p_{s,j}^{n+1} = 0. \tag{39}$$

For body j: the kinematic matching condition (**??**),

$$\frac{\phi_{n_z-1,j}^{n+1} - \phi_{n_z-2,j}^{n+1}}{\Delta z} + w_{n_z-1,j}^{3,n+1} - v_b^{n+1} = 0. \tag{40}$$

These rows serve dual purposes: pinning the pressure to zero outside the body, and enforcing the no-penetration constraint (with p_s as the Lagrange multiplier) inside.

Surface dynamic BC (uniform). The dynamic condition (**??**) is discretised uniformly for all surface nodes j, with the p_s term always present:

$$\left(I - \frac{2\Delta t}{\mathrm{Re}}\,\Delta_H\right)\phi^{n+1}_{n_z-1,j} + \frac{2\Delta t}{\mathrm{Re}}\,\frac{w^{3,n+1}_{n_z-1,j} - w^{3,n+1}_{n_z-2,j}}{\Delta z} + \Delta t\, p^{n+1}_{s,j} = \phi^{n}_{n_z-1,j} - \frac{\Delta t}{\mathrm{Fr}}\eta^n_j + \frac{\Delta t}{\mathrm{We}}\kappa[\eta^n]_j. \tag{41}$$

For free j, $p^{n+1}_{s,j} = 0$ is enforced by the separate p_s row above, so this reduces to (**??**).

3.4.3 Body dynamics rows

Implicit Euler discretisation of (**??**)–(**??**) yields two additional rows:

$$z^{n+1}_b - \Delta t\, v^{n+1}_b = z^n_b, \tag{42}$$

$$M\, v^{n+1}_b - \Delta t\, \Delta x \sum_{j\in\mathcal{B}} p^{n+1}_{s,j} = M\, v^n_b - \frac{\Delta t\, M}{\mathrm{Fr}}. \tag{43}$$

The sum in (**??**) is the midpoint approximation to the integrated fluid pressure over the wetted region. These two rows close the augmented system.

3.4.4 Block structure and matrix properties

The augmented matrix retains a block lower-triangular structure with respect to the main dependencies, with off-diagonal coupling blocks between ϕ and $\{w^3, p_s, \eta\}$, and between v_b and p_s (via the pressure integral). Crucially, all coefficients are determined by grid parameters, Δt, the dimensionless numbers (Fr, We, Re), the body parameters (M, a, x_c), and the time-independent body mask χ. None depend on the solution. Therefore, the augmented matrix is factored once via sparse LU decomposition at setup time, and each subsequent time step reduces to a single back-substitution. The augmented system has $N_{\mathrm{total}} = 3\, n_z n_x + 2\, n_x + 2$ unknowns, increased only by $n_x + 2$ compared with the pure fluid system.

4 Results

5 Discussion

6 Conclusions